Effective: 22 March 2026

LATIN AMERICAN SPACE CHALLENGE

Lander Challenge
Standards Manual (LCSM)

Space Challenge

Edition 1 | Revision 1

The electronic version is the official, approved document.
Ensure you are using the correct version before proceeding.

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LATIN AMERICAN SPACE CHALLENGE

Table of Contents

Table of Contents 2
Record of Revisions 3
LCSM First Edition 4
Introduction 5
1. Purpose 5
2. Structure 5
3. LASC Standards and Recommended Practices (LSARPs) 5
4. Manual Revisions 6
5. Acronyms and Definitions 6
Section 1 — Lander Challenge (LND) 9
1.1. Challenge Overview 9
1.2. Mission Categories 10
1.3. Milestones 10
1.4. Challenge Deliverables 15
1.5. Scores, Evaluation and Awards 20
1.6. Specific Awards 21
Section 2 — Safety (SAF) 22
2.1. Safety Policy 22
2.2. Mandatory Safety Guidance 22
2.3. Stored-Energy Devices 23
2.4. Pressure Vessels 25
Section 3 — Propulsion System (PRS) 27
3.1. Guidance 27
3.2. Propellants 27
3.3. Engineering Recommendations and Testing 27
3.4. Propulsion Safing and Arming 28
Section 4 — Airframe and Aerodynamics (STR) 30
4.1. Overall Structural Integrity 30
4.2. Material Selection and Tests 30
4.3. Best Practices and Recommendations 30
Section 5 — Electronics and Control Systems (ECS) 32
5.1. Official Altitude Logging System 32
5.2. Power Systems and Critical Wiring 32
5.3. Attitude Control Systems (ACS) 35
Section 6 — Hot-fire Test Operations (HFT) 36
Section 7 — Launch Operations (FLT) 37
Section 8 — General Requirements (GEN) 38
8.1. Administrative Requirements 38

Appendix A: Details for the Milestone Report **39**
Appendix B: Detailing Grading Criteria **42**

Record of Revisions

The table below records all revisions of the Lander Challenge Standards Manual (LCSM) since its initial release for the Latin American Space Challenge.

Edition Number	Revision Number	Issue Date	Effective Date
1	1	March 2026	March 2026

LCSM First Edition

The following table describes changes contained in Edition 1 of the Lander Challenge Standards Manual (LCSM). The table describes significant changes and individual changes in each of the LCSM sections and provisions.

LCSM 1 Revision 1 Highlights	
Area(s) of Change	**Description(s)**
All Document.	Initial release of the document.

Introduction

1. Purpose

The Latin American Space Challenge (LASC) Lander Challenge Standards Manual (LCSM) is published to establish the standards, guidelines, and protocols necessary for participants to effectively engage in the LASC Lander Challenge.

This manual serves as the definitive reference for teams preparing and executing missions within the competition. It defines the criteria, requirements, and best practices to be followed throughout the lifecycle of the lander project, including design, development, and, where applicable, assembly, integration, testing, and low-altitude/low-velocity operations. It also provides the framework used to evaluate and assess the technical and operational aspects of all entries, ensuring alignment with best practices in space engineering and mission planning.

2. Structure

The LCSM is based on the LASC Documentation System described in Section 2 of the LASC General Rules (LGR) document. This documentation system is structured into three hierarchical levels of documents that establish the overall framework of the event and its challenges.

Adherence to this documentation system by teams and participants is mandatory in order to ensure safety, organization, and compliance with event requirements.

The LCSM is organized as follows

- Section 1 → Lander Challenge (SAT);
- Section 2 → Safety (SAF);
- Section 3 → Propulsion System (PRS);
- Section 4 → Airframe and Aerodynamics (STR);
- Section 5 → Electronic Power System and Communications (ECS);
- Section 6 → Hot-fire Test Operations (HFT);
- Section 7 → Launch Operations (FLT);
- Section 8 → General Requirements (GEN);

Each section in this Manual is assigned an associated 3-letter identifier (in parentheses above). The reference number for every standard or recommended practice within a section includes the specific 3-letter identifier for that section (e.g., LND 1.1.1).

3. LASC Standards and Recommended Practices (LSARPs)

LSARPs in this manual are designed for use within the Latin American Space Challenge (LASC) and provide the criteria for evaluations. The LASC General Rules (LGR) document defines the Documentation System and LSARPs.

4. Manual Revisions

The document is subject to an annual revision, which invariably leads to a new edition. In the event that critical issues emerge impacting the content of the document, a revision to the current edition will be undertaken. All changes in this document are listed in the revision highlights table. For easier orientation, the following symbols identify any changes made within each section:

- △ Addition of a new item.
- ↻ Change to an item.
- ⊗ Deletion of an item.

5. Acronyms and Definitions

AGL - Above Ground Level: The altitude of the rocket measured vertically from the launch pad ground elevation to the vehicle position or apogee.

All altitude requirements in the Rocket Challenge shall be measured in meters Above Ground Level (AGL) from the launch pad elevation. GPS receivers typically measure altitude relative to Mean Sea Level (MSL), therefore teams shall ensure proper conversion when reporting altitude data.

Caliber: A unit of length equal to the external diameter of the airframe body tube.

CG - Center of Gravity: The point at which the total mass of the vehicle is considered to be concentrated. The vehicle will balance at this point in any orientation.

CP - Center of Pressure: The point along the vehicle where the resultant aerodynamic forces are considered to act. The relative position between the Center of Pressure and the Center of Gravity determines the aerodynamic stability of the vehicle.

EGM - Event Guidance and Milestones: LASC Event Guidance and Milestones document available in the Documentation section of the LASC website.

LCC - Launch Control Center: The designated area from which launch operations are monitored and controlled, including vehicle status monitoring, safety coordination, and launch authorization.

LGR - LASC General Rules: LASC General Rules document available in the Documentation section of the LASC website.

LRR - Launch Readiness Review: A formal review conducted to verify that the vehicle, team, and operations are ready for launch.

LSARP - LASC Standards and Recommended Practices: Standards and recommended practices defined and detailed in the LASC General Rules (LGR) document.

MCC - Mission Control Center: The designated area from which mission operations are monitored and controlled after launch, including telemetry monitoring, flight tracking, command operations, and mission status management.

MSL - Mean Sea Level: Altitude measured relative to the average sea level, commonly used in aviation and GPS altitude measurements.

Payload: A single mass that does not interact with the overall operation of the vehicle. The mass must be a single piece (ie, not distributed in small pieces around the vehicle) and cannot be dual purpose with respect to vehicle operation.

Propulsion System: A single liquid or hybrid engine that provides the primary vertical lifting force of the vehicle.

No aerodynamic or air-breathing methods are permitted (ie, jets or fans). However, thrusters, solid motors and other in-space-type propulsion systems are permitted for control other than the primary lifting force (ie, attitude, roll, etc), except for in the TVC or Throttle Milestones (these other thrust sources are not considered part of the Propulsion System).

The intent is that all milestones use a full-scale propulsion system, as opposed to a subscale one. Event officials understand teams will iterate heavily, and possibly scrap, their engines as they progress – it is fine for the hop engine to perhaps be quite different from the engine used in the TVC milestone.

RAA - Rocket Assembly Area: Designated area where teams assemble, integrate, and prepare their rockets for flight.

RBF - Remove Before Flight: Safety tag or device that must be removed prior to launch or flight operations.

SF - Safety Factor: The ratio between the maximum load or stress a component can withstand and the maximum load or stress expected during operation. The safety factor is used to provide a margin against uncertainties, material variability, and unexpected loads.

Selected Mission: A mission proposal submitted by a team that has been evaluated, approved, and officially accepted by the LASC Organization for participation in the event.

SMS - Safety Management System: A systematic approach to managing safety, including organizational structure, responsibilities, policies, procedures, risk management, and continuous safety improvement processes.

R&D — Researched and Developed: A designation for hardware, software, or systems that have been primarily designed, developed, and built by the team, rather than purchased as complete commercial off-the-shelf solutions.

Static Stability Margin: The distance between the Center of Gravity and the Center of Pressure divided by the airframe diameter, expressed in calibers.

Successful Landing: The vehicle must land gently VTVL style (ie, a vertical propulsive landing as opposed to a horizontal "airplane" landing), and afterwards be in good condition, such that one may reasonably consider reusing the vehicle again with minor repair. Designed-to-fail landing systems, like crush structure legs, are not permitted. The entire vehicle that took off must land.

Tether: A physical restraint that prevents the vehicle from deviating substantially from a nominal flight area under the worst case scenario (i.e. throttle stuck open). The tether shall not assist the vehicle or substantially influence its flight during a nominal test (ie, while hovering).

Section 1 — Lander Challenge (LND)

Section 1 defines the Lander Challenge (LND) mission categories, establishes the administrative requirements for application and registration, outlines all technical deliverables, and details the evaluation structure and awards.

1.1. Challenge Overview

The Lander Challenge is a long-term technical initiative structured around annual objectives for missions focused on the development of self-landing space vehicles. The initiative recognizes teams that achieve significant milestones toward the development of reusable launch systems or spacecraft capable of autonomous vertical landing.

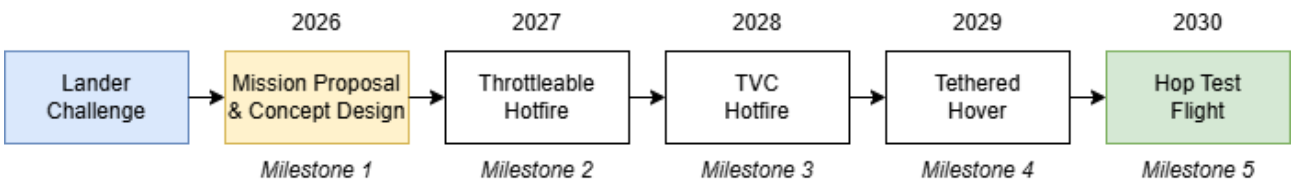
The primary objective of the Lander Challenge is to enable and inspire teams to pursue the development of reusable, self-landing launch vehicles or planetary landers, while fostering expertise across the engineering and operational disciplines required for such systems.

Teams should note that the challenge does not necessarily require large-scale propulsion systems. For example, performing a 50-meter hop test does not inherently require a powerful engine or a full Thrust Vector Control (TVC) system. The challenge is designed to support incremental development approaches, serving as a stepping stone toward more advanced aerospace projects.

Yearly Milestones

The challenge is structured around five major milestones, as illustrated in Figure 1, to be achieved within a period of up to five years, depending on the mission's development progress. Participation requires teams to actively develop a self-landing vehicle concept as the core element of their project.

Figure 1: Yearly Milestones.



Each year, the Latin American Space Challenge (LASC) opens an application process for new missions interested in joining the challenge. All applicable rules and requirements are defined within the LASC Documentation System and may be updated or amended by LASC at any time.

For the 2026 edition, the challenge will focus on the Mission Proposal and Concept Design phases of the Milestone 1. Participating teams are required to timely submit mission proposals,

development documentation, including progress updates, a mission patch, a pitch video, Milestone Report and presentation materials.

Starting in 2027, the organization intends to provide operational testing opportunities during the onsite event, enabling participating missions to test throttleable propulsion systems.

1.2. Mission Categories

Teams intending to participate in the LanderChallenge with a mission shall submit an Entry Application as defined in the LASC General Rules (LGR), indicating the desired mission category.

LND 1.2.1. Missions competing in the Lander Challenge shall develop a lander project under one of the following mission categories:

- Reusable Launch System (RLS) project; or
- Planetary Lander (Moon or Mars) project

The Lander Challenge is structured around five milestones for each mission category during a multi-year roadmap. These milestones follow a similar structure but include different technical requirements depending on the mission category.

The milestones serve as a development roadmap toward the primary objective of building advanced capabilities in complex aerospace systems for teams participating in the Latin American Space Challenge.

1.3. Milestones

Milestone 1 – Mission Proposal & Concept Design

Milestone 1 focuses on mission definition and system-level concept development. Participating teams with selected missions are expected to develop a mission proposal extending through 2030 and covering all planned milestones.

LND 1.3.1. Each mission shall define a mission concept for a self-landing vehicle corresponding to the selected mission category.

LND 1.3.2. Each mission shall conduct and present a Concept Design Review (CoDR) of the proposed system.

LND 1.3.3. The Concept Design Review (CoDR) shall present the overall system architecture, development approach, and programmatic planning for the mission.

LND 1.3.4. The Concept Design Review shall include, at a minimum, the following elements:

a) System architecture and technical concept;
b) Development budget estimation;
c) Proposed project schedule and development timeline;
d) Preliminary risk analysis.

LND 1.3.5. The Mission Proposal and Concept Design Review shall demonstrate the technical feasibility and development approach of the proposed mission.

LND 1.3.6. All documentation shall be submitted in accordance with the LASC Documentation System requirements established for the competition.

There are no category-specific technical performance requirements for Milestone 1 beyond the development of the Mission Proposal and Concept Design for the selected missions.

Milestone 2 – Throttleable Hotfire

General Requirements

LND 1.3.7. The Milestone 2 — Throttleable Hotfire requires each selected mission to perform, during the onsite event, a hot-fire test of a propulsion system with throttleable capabilities.

LND 1.3.8. The propulsion system shall demonstrate controlled thrust modulation during a continuous hot-fire test.

LND 1.3.9. Thrust modulation shall be achieved by controlling the propulsion system itself. The use of auxiliary systems such as gas thrusters to artificially modify thrust levels shall not be permitted for this milestone.

LND 1.3.10. Each mission shall submit a planned thrust trace prior to the hot-fire attempt.

LND 1.3.11. Each mission shall provide load cell data, control system data, and any other relevant telemetry demonstrating compliance with all requirements. Data shall be presented in a clear and organized format for evaluation by the judges.

LND 1.3.12. During throttle-down and throttle-up transitions, an additional tolerance window of 0.5 seconds shall be allowed for timing compliance

Reusable Launch System (RLS) Requirements

LND 1.3.13. For Reusable Launch System (RLS) missions, the propulsion system shall operate for a minimum of 10 seconds of controlled thrust, excluding transient startup periods.

LND 1.3.14. The propulsion system shall demonstrate controlled and constant nominal thrust between 2 kN and 5 kN for a minimum duration of 5 seconds.

LND 1.3.15. The propulsion system shall smoothly throttle down to a thrust level within the range of 40% to 60% of the nominal thrust and shall maintain a constant thrust within this range for not less than 3 seconds.

LND 1.3.16. The propulsion system shall return to the same nominal thrust level and shall maintain constant thrust within the range of 2 kN to 5 kN for not less than 2 seconds.

LND 1.3.17. The planned thrust trace shall include an upper and lower tolerance band of ±10% of the planned nominal thrust.

LND 1.3.18. The measured thrust shall remain within the ±10% tolerance band centered on the planned nominal thrust throughout the thrust profile.

LND 1.3.19. During the nominal thrust hold period, the measured thrust shall remain above 1.8 kN for the entire duration.

LND 1.3.20. The low-thrust reference range (40% to 60%) shall be determined based on the lowest measured thrust value recorded during the nominal thrust hold period.

LND 1.3.21. During the low-thrust hold period, the measured thrust shall remain between 40% and 60% of the lowest thrust value measured during the nominal thrust hold period.

Planetary Lander (Moon or Mars) Requirements

LND 1.3.22. For Reusable Launch System (RLS) missions, the propulsion system shall operate for a minimum of 20 seconds of controlled thrust, excluding transient startup periods.

LND 1.3.23. The propulsion system shall demonstrate controlled and constant nominal thrust between 0.8 kN and 2 kN for a minimum duration of 8 seconds.

LND 1.3.24. The propulsion system shall smoothly throttle down to a thrust level within the range of 35% to 45% of the nominal thrust and shall maintain a constant thrust within this range for not less than 4 seconds.

LND 1.3.25. The propulsion system shall return to the same nominal thrust level and shall maintain constant thrust within the range of 0.8 kN and 2 kN for not less than 8 seconds.

LND 1.3.26. The planned thrust trace shall include an upper and lower tolerance band of ±5% of the planned nominal thrust.

LND 1.3.27. The measured thrust shall remain within the ±5% tolerance band centered on the planned nominal thrust throughout the thrust profile.

LND 1.3.28. During the nominal thrust hold period, the measured thrust shall remain above 0.8 kN for the entire duration.

LND 1.3.29. The low-thrust reference range (35% to 45%) shall be determined based on the lowest measured thrust value recorded during the nominal thrust hold period.

LND 1.3.30. During the low-thrust hold period, the measured thrust shall remain between 35% and 45% of the lowest thrust value measured during the nominal thrust hold period.

Milestone 3 – TVC Hotfire

General Requirements

LND 1.3.31. During the TVC hot-fire test, the propulsion system shall demonstrate thrust vector control by deflecting the thrust vector by a minimum angle of 7 (seven) degrees in all directions in a controlled manner, as determined acceptable by the judges.

LND 1.3.32. Thrust vector control shall be achieved by manipulation of the primary propulsion system. The use of external gas thrusters or reaction control thrusters to change the thrust vector during this milestone shall not be permitted.

LND 1.3.33. Alternative thrust vector control methods such as flow impingement, jet vanes, movable nozzles, or gimbaled engines shall be permitted.

LND 1.3.34. Vehicles may use gas thrusters or other attitude control systems during hover, drop, or hop tests in later milestones; however, such systems shall not be used to accomplish the TVC requirement of this milestone.

LND 1.3.35. Load cell data, control system data, actuator position data, and any other relevant telemetry demonstrating compliance with the thrust vector control requirements shall be provided in a clear and organized format for evaluation by the judges.

Reusable Launch System (RLS) Requirements

LND 1.3.36. For Reusable Launch System (RLS) missions, the propulsion system shall operate within a thrust range of 2 kN to 5 kN for a minimum duration of 10 seconds during the TVC hot-fire test.

Planetary Lander (Moon or Mars) Requirements

LND 1.3.37. For Planetary Lander missions, the propulsion system shall operate within a thrust range of 0.8 kN to 2 kN for a minimum duration of 20 seconds during the TVC hot-fire test.

Milestone 4 – Tethered Hover

The Milestone 4 — Tethered Hover requires the vehicle to perform a controlled hover flight while attached to a tether system during the test.

General Requirements

LND 1.3.38. The tether system shall be used as a safety restraint and shall not be used to actively control or stabilize the vehicle during the hover test.

LND 1.3.39. The vehicle shall demonstrate controlled hover flight using its propulsion and control systems.

LND 1.3.40. Each mission shall provide avionics data, acceleration data, altitude data, and any other relevant telemetry demonstrating compliance with all requirements. Data shall be presented in a clear and organized format for evaluation by the judges.

LND 1.3.41. For the purposes of this milestone, controlled hover shall be defined as maintaining vehicle position within a defined hover control volume during the hover period.

LND 1.3.42. Unless otherwise approved by the judges, the vehicle centroid shall remain within a 1-meter cubic volume during the hover test.

LND 1.3.43. The hover timer shall start when the vehicle first achieves stable hover and shall end when the vehicle leaves the hover control volume, lands, or thrust is reduced below hover thrust.

Reusable Launch System (RLS) Requirements

LND 1.3.44. For Reusable Launch System (RLS) missions, the vehicle shall perform a controlled tethered hover for a minimum duration of 10 seconds while carrying at least 15 kg of payload mass.

Planetary Lander (Moon or Mars) Requirements

LND 1.3.45. For Planetary Lander missions, the vehicle shall perform a controlled tethered hover for a minimum duration of 20 seconds while carrying at least 5 kg of payload mass.

Milestone 5 – Hop Test Flight

The Milestone 5 — Hop Test Flight requires the vehicle to perform a controlled ascent, descent, and landing using the primary propulsion system while carrying the declared payload mass.

General Requirements

LND 1.3.46. The vehicle shall take off from the ground, ascend to an altitude of 50 meters above ground level (AGL), and perform a controlled descent followed by a successful landing.

LND 1.3.47. The landing shall occur within a landing circle with a diameter of 10 meters located at a position selected by the judges.

LND 1.3.48. For a landing to be considered successful, the entire vehicle shall be contained within the vertical projection of the landing circle at the time of landing, as determined by the judges.

LND 1.3.49. Minor detachable or negligible materials such as thermal insulation fragments or small debris shall not be considered part of the vehicle for landing boundary determination, at the judges' discretion.

LND 1.3.50. The vehicle shall demonstrate controlled flight during ascent, hover (if performed), descent, and landing, as determined acceptable by the judges.

LND 1.3.51. Each mission shall provide avionics data, GNSS data, acceleration data, altitude data, and any other relevant telemetry demonstrating compliance with all requirements. Data shall be presented in a clear and organized format for evaluation.

LND 1.3.52. The vehicle shall include a flight termination system, kill switch, or equivalent safety system capable of shutting down the propulsion system in case of emergency.

LND 1.3.53. Teams shall follow safe operational practices and all safety requirements established for the event.

Reusable Launch System (RLS) Requirements

LND 1.3.54. For Reusable Launch System (RLS) missions, the vehicle shall perform a controlled tethered hover for a minimum duration of 10 seconds while carrying a payload mass of not less than 15 kg.

LND 1.3.55. The vehicle shall take off and land within the same launch pad area.

LND 1.3.56. The total flight duration shall be not less than 10 seconds.

Planetary Lander (Moon or Mars) Requirements

LND 1.3.57. For Planetary Lander missions, the vehicle shall perform a controlled tethered hover for a minimum duration of 20 seconds while carrying a payload mass of not less than 5 kg.

LND 1.3.58. The vehicle shall take off from a designated launch pad and land at a designated landing zone located at a distance of not less than 150 meters from the launch pad.

LND 1.3.59. The total flight duration shall be not less than 20 seconds.

1.4. Challenge Deliverables

The following sections define the deliverable materials event officials require from teams competing in the Lander Challenge, including each deliverable's format and content.

All deliverables will be submitted to LASC per the instructions provided to the teams. Each relevant deliverable description will facilitate submission of that deliverable or will be communicated to teams as is determined by LASC Organization.

The scheduled due dates of all required deliverables are recorded in the EGM document, maintained on the LASC website.

Progress Updates

There will be four Progress Updates submitted per Selected Mission due to track mission progress in design and development. These forms will contain questions regarding safety, organization, launch operations and testing, propulsion, airframe and structures, avionics, recovery, and payload.

The template for each progress update will be available at least two weeks before the deadline to allow teams sufficient time to collect the appropriate information and data.

LND 1.4.1. Teams shall submit Progress Updates for each Selected Mission via the Latin American Space Challenge website (https://www.lasc.space/) on 4 (four) specific occasions prior to the onsite event: the Progress Update 1, Progress Update 2, Progress Update 3 and the Progress Update 4.

These Progress Updates will record progression in the project's technical characteristics during development. Event officials understand not all technical details will be known until later in the design process. Therefore, the Progress Updates prior to the final submission will be evaluated based only on their timeliness and completeness.

LND 1.4.2. Total completeness of the Progress Update form is required at all times. Reasonable engineering estimates and approximations are expected during the application process, but will be subject to progressive additional scrutiny in the subsequent Progress Updates.

Each mission should briefly mention their ongoing discussions and analysis in the comment fields for any numerical submissions that are known to be unreasonable or remain undecided.

LND 1.4.3. Teams should briefly mention their ongoing discussions and analysis in the comment fields for any numerical submissions that are known to be unreasonable or remain undecided.

Teams may also respond to undecided criteria by demonstrating their understanding of any applicable event guidance or best practice governing the particular detail.

Marketing Actions

Since the early days of the Apollo missions, the general public has been captivated by the spectacle of rocket launches, eager for more information and the chance to witness these events firsthand. The allure of a launch vehicle is immense, drawing thousands of people to travel far from home to be near launch sites, just to hear the thunderous roar of rocket engines.

Event officials recognize the importance of structured marketing efforts to promote each team and its members. These efforts aim to increase visibility, attract attention to the missions, and help

support fundraising for each project. Such efforts are essential to realizing the dream of having thousands of spectators watch experimental rockets and satellites launch from LASC.

The Marketing Actions are divided in three deliverables: a Mission Patch, a Pitch Video and a competition for likes/followers in the following social media: Instagram, TikTok, X (Twitter), Facebook and LinkedIn.

Mission Patch

Mission patches are emblems designed and worn by astronauts and people affiliated with a mission, such as Rocketeers and Satelliteers. The patches depict an image associated with the mission.

LND 1.4.4. Each team shall submit an appropriate Mission Patch for each Selected Mission during the Progress Update 1. Submissions shall be made in a timely manner using the designated form on the LASC Website, following the schedule outlined in the deliveries roadmap.

A Mission Patch must contain the mission and event references. The Mission Patch must include, at a minimum:

- The Mission ID and mission reference;
- The event reference;
- The milestone reference; and
- The year of the event.

For more information and examples, teams should access the following websites:

- Wikipedia - Mission Patch: https://en.wikipedia.org/wiki/Mission_patch
- Human Spaceflight Mission Patches: https://www.nasa.gov/gallery/human-spaceflight-mission-patches/
- NASA - Houston We Have a Podcast, Mission Patches: https://www.nasa.gov/podcasts/houston-we-have-a-podcast/mission-patches/
- National Air and Space Museum - The Meaning of Mission Patches: https://airandspace.si.edu/stories/editorial/meaning-mission-patches
- SpaceX Unofficial Patch List: https://spacexpatchlist.space/
- Blue Origin's NS-30 Mission Patch: https://www.instagram.com/p/DGb0jnkNspS/

The team's logo will not be accepted as a mission patch. Late submission will be accepted, but the mission may be penalized by not timely submitting a Mission Patch.

Pitch Video

Reels, short videos, and pitch presentations—these are new terms that have become part of our daily lives. The younger generations are completely immersed in social media, especially videos: easy to watch and fun! To make the competition more diverse and not just about technical

development, it is crucial to communicate what each selected mission is doing to the broader society. Therefore, each selected mission will be required to record a pitch video explaining their goals, concept of operations, and anything else the team feels needs to be communicated.

LND 1.4.5. Each selected mission shall submit a Pitch Video explaining the mission proposal, goals, concept of operations, and other important information and data regarding the concept design of the lander project, including budget, roadmap and schedule.

LND 1.4.6. The Pitch Video shall have a minimum duration of 2 minutes and a maximum duration of 5 minutes.

LND 1.4.7. The Pitch Video shall have clear audio and video quality (720p minimum) sufficient for proper evaluation.

LND 1.4.8. The Pitch Video should include images, videos, animations, or footage of the team, the project, hardware, simulations, or development activities.

LND 1.4.9. On or before a specified date prior to the event, teams shall submit the YouTube link of the video using the appropriate location indicated on LASC website.

The video shall be uploaded on YouTube with a title as “Mission Your_Mission_ID Pitch Video to the Event Edition LASC". For example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their YouTube Video as "Mission 19 Pitch Video to the 7th LASC".

Milestone Report (MR)

Each team shall submit a Milestone Report with a mission proposal describing their experimental lander project, concept of operations, budget, and roadmap to the technical evaluation board and event officials. This document shall include detailed information on the design review, goals, tests, and the most accurate technical data of the vehicle.

LND 1.4.10. Teams shall timely submit a digital, PDF copy of their Milestone Report for each Selected Mission, with the file name "Your Mission ID_Milestone Report". For example, a team assigned the Mission ID "19" would submit a digital copy of their Milestone Report using the filename "19_Milestone Report".

LND 1.4.11. The Milestone Report shall be formatted according to the style guide of the Latin American Space Challenge (LASC), using a provided Microsoft® Word document template. Always check the template maintained on the website before drafting your Milestone Report to ensure you are using the latest version.

LND 1.4.12. The Milestone Report shall be no longer than 50 pages, including figures, footnotes, sources, source endnotes, nomenclature lists, equations, explanations of variables etc. This does include the Appendices. However, appendices are not necessarily read in detail by the event officials.

LND 1.4.13. The Milestone Report's main title is left to the team's discretion, however; the paper shall be subtitled “Milestone Report *Your Mission* ID to the Event Edition Latin American Space Challenge".

For example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their Milestone Report "Milestone Report 19 to the 7th Latin American Space Challenge".

Further requirements are given in Appendix A: Details for the Milestone Report, including the required minimum Milestone Report sections and appendices. Additional sections, subsections, and appendices may be added as needed.

Presentation Session

Event officials aim to promote networking and knowledge sharing between teams during the onsite event. Presentation Sessions will be held in designated slots to foster an environment of technology development.

LND 1.4.14. Selected lander missions shall register to participate as speakers in a Presentation Session, discussing the proposed mission.

The date of the official announcement of selected missions chosen to speak in the Presentation Session will be published in the LASC Event Guidance and Milestones (EGM) document.

LND 1.4.15. Each mission shall submit a Presentation Session Abstract on their work for competition officials and the judging panel to consider including in a Presentation Session held during the on-site event. The file name should be "Your Mission ID_Presentation Session Slides." For example, a team assigned the Mission ID "19" would submit their slides using the filename "19_Presentation Session Abstract".

LND 1.4.16. Each mission shall submit a digital PDF copy of their Presentation Session Slides for a specific Selected Mission to the appropriate LASC submission form available on the website.

The file name should be "Your Mission ID_Presentation Session Slides." For example, a team assigned the Mission ID "19" would submit their slides using the filename "19_Presentation Session Slides."

LND 1.4.17. The Presentation Session Abstract shall follow the formatting guidelines provided in the LASC Presentation Abstract Template, available on the LASC website. There is no template for the Presentation Session Slides in order to preserve each team’s creativity.

However, all slides must include the Official LASC Event Patch in one of the corners to clearly identify the event in the presentation materials.

LND 1.4.18. The Presentation Material's main title is left to the team's discretion. However the first slide shall be subtitled “Presentation Session Your Mission ID to the Edition LASC". For

example, a team assigned the Mission ID "19", competing in the 7th LASC, would subtitle their Presentation Slide “Presentation Session for Mission 19 to the 7th LASC".

LND 1.4.19. The Presentation Material can be given in English, Portuguese, or Spanish; however, the slides shall be in English. Teams should inform the language to be used during the Presentation Session in the appropriate LASC submission form. The language shall not be changed after submission.

LND 1.4.20. Each Presentation Session shall last between 7 and 10 minutes for the presentation, followed by up to 5 minutes for questions and answers, totaling a maximum session duration of 15 minutes.

LASC Officials will review all submissions and select the missions invited to present during the Presentation Session. The selected missions will be announced in a livestream at least one week before the on-site event.

1.5. Scores, Evaluation and Awards

Event officials will evaluate competitors for the Milestone Award within each mission category based on the quality of required project documentation, the quality of their system’s overall design and simulation, and the mission’s overall excellence demonstrated at the Milestone Report.

Scoring Categories and Grading Criteria

In 2026, lander missions will be scored in five different scoring categories or areas, which are (1) Team Effort, (2) the Milestone Report, and (3) the Presentation Session. These are weighted according to the following table. In each scoring category, a set of grading criteria is established. These criteria will be evaluated by the jury for each mission individually. Each grading criterion has several, more detailed, topics that establish what the jury will look for during the grading process.

Table 2: Weight of the Scoring Categories for the Rocket Challenge.

Scoring Categories	Possible Points	% of Total Points
Team Effort	250	25%
Milestone Report	400	40%
Presentation Session	350	35%
TOTAL	**1000**	**100%**

These detailed topics are weighed equally within each criterion, while the main criteria are weighted differently within each mission category. The details of the grading criteria can be found in Appendix B: Detailed Grading Criteria.

1.6. Specific Awards

The lander Challenge awards the Milestone Winner in each category. Overall Awards are defined in the LASC General Rules (LGR) document.

Best Experimental Lander of the Year

One team among the Milestone Winner Award in each category defined in this document will be named the Best Experimental Lander of the Year. The recipient of this award is determined by qualitative assessments of the event officials made throughout the entire event.

The Best Experimental Lander of the Year not necessarily will be the highest scoring mission, but will be awarded to a single team chosen by the event officials based on project complexity, technology and innovation.

A team is considered eligible for the award after participating in the Lander Challenge submitting all documents, reports, activities, and presenting the project.

Milestone Winner Awards

A Milestone Winner Award will be granted to the highest scoring, eligible team in each of the Mission Categories defined in this document. A team is considered eligible for the place award(s) in its category after participating in the 2026 LASC submitting all documents, reports and activities.

In the event, if no teams meet this definition in a given category, event officials may issue Milestone Winner Awards at their discretion based on multiple factors – including points accrued, participation and engagement, and overall performance.

Section 2 — Safety (SAF)

Section 3 details the Safety (SAF) policy and provisions for all missions, members, and teams to ensure event safety. This section covers the general guidelines for Satellite Missions.

2.1. Safety Policy

Safety is our Value number 1.

The Safety Management System (SMS) of the Latin American Space Challenge is composed of four pillars: Safety Policy, Safety Risk Management, Safety Assurance, and Safety Promotion.

All team members shall report any observed unsafe activities to event officials. The event will provide a specific channel for these reports, which will be anonymized before any potential action is taken by event officials.

The event will be guided by a Just Culture approach, where errors will be addressed through reeducation. However, any violation of rules will be treated as potential unruly behavior, with penalties or disqualification possible as outlined in Section 2.

During the Launch Readiness Review (LRR), event officials will use a checklist to approve an experimental rocket for loading with energetics and a rocket motor. Ensuring safety will be a key process for issuing any Flight Card during the event.

Event officials will conduct a safety risk management analysis before each launch to reduce the potential probabilities and/or severities of any operational risks.

Promotion is key to raising the bar among all participating teams. Event officials will conduct safety briefings to increase overall public awareness. Launching an experimental rocket is not trivial; the event has associated risks. All participants shall understand and accept these risks by signing a Waiver and Release of Liability to participate.

SAF 2.1.1. All participants shall follow the Latin American Space Challenge Safety Policy.

SAF 2.1.2. Participants in the Lander Challenge - Milestone 1 will not engage in hands-on activities, but must adhere to the Safety Policy at all times. The Mandatory Safety Guidance shall be followed during the experimental lander concept proposal.

2.2. Mandatory Safety Guidance

SAF 2.2.1. No energetics can be stored and/or transported in closed vessels or any other condition that may cause pressurization. Therefore, propellant grains shall be kept outside of casings and stored in proper containers.

SAF 2.2.2. All propulsion systems shall be inspected during the Hot-fire Test Readiness Review (HTRR) or Launch Readiness Review (LRR).

SAF 2.2.3. In case of hybrid propulsion systems, if the grain is made of a single segment, and is no longer than 600 mm, it may be inspected inside the combustion chamber. However, it is highly recommended that all propellant grains be outside for a complete inspection during the HTRR or LRR.

If the team (hybrids only), for any reason, chooses to place the grain inside the combustion chamber prior to inspection, it is a good practice to take pictures and videos of the grain and its installation. This may aid the officials to validate the structural integrity.

SAF 2.2.4. All heaters and igniters for propulsion systems shall be inspected during the HTRR or LRR before authorization for final assembly.

SAF 2.2.5. In case of liquid and/or hybrid propulsion systems, propellant tanks only will be loaded on the test bench or launch base with proper authorization from event officials and their supervision.

SAF 2.2.6. Teams with selected hybrid/liquid propulsion missions shall discuss their systems with their Rocketry Mentor, who will be responsible for contacting LASC to schedule a meeting. During this meeting, the team will present their system (including propellants, propellant masses, tanking configuration, feed system, and filling system) alongside their RM.

The RM shall schedule this meeting via email to lasc@lasc.space. During the meeting, event officials will define a safety distance for propellant filling and hose removal.

SAF 2.2.7. The personnel involved in launch operations and energetics integration shall wear Personal Protective Equipment (PPE). The minimum PPE required are the helmet and a signaling vest, temporarily provided by the event during the launch preparations. Additional PPE provided by the teams are welcome.

SAF 2.2.8. The personnel involved in launch operations and energetics integration shall wear full-length pants and closed shoes.

2.3. Stored-Energy Devices

Energetic Device Safing and Arming

An energetic device is considered armed when only one event is necessary to release the energy. For the purpose of this document, energetics are defined as all stored-energy devices - other than propulsion systems - that have reasonable potential to cause bodily injury upon energy release.

The following table lists some common types of stored-energy devices and overviews in what configuration they are considered non-energetic, safed, or armed.

Table 3: Energetic Devices and Configurations.

Device Class	Non-energetic	Safed	Armed
Igniters/Squibs	Small igniters/squibs, nichrome, wire or similar	Large igniters with leads shunted	Large igniters with no shunted leads
Pyrogens (e.g., black powder)	Very small quantities contained in non-shrapnel producing devices (e.g., pyro-cutters or pyro-valves)	Large quantities with no igniter, shunted igniter leads, or igniter(s) connected to unpowered avionics	Large quantities with non-shunted igniter or igniter(s) connected to powered avionics
Mechanical Devices (e.g., powerful springs)	De-energized/relaxed state, small devices, or captured devices (i.e., no jettisoned parts)	Mechanically locked and not releasable by a single event	Unlocked and releasable by a single event
Pressure Vessels	Non-charged pressure vessels	Charged vessels with two events required to open main valve	Charged vessels with one event required to open main valve

Although these definitions are consistent with the propulsion system arming definition, this requirement is directed mainly at the energetics used by the termination system as well.

Note that propulsion systems may be armed only after the launch pad area is evacuated to a specified distance, this requirement permits personnel to arm other stored-energy devices at the launch rail.

SAF 2.3.1. All energetics shall be in the safe position/safed (i.e., “remove before flight connected”) until the experimental rocket is in the launch position, at which point they may be "armed". An energetic device is considered in the safe position/safed when two separate events are necessary to release the energy.

SAF 2.3.2. All energetic device arming features shall be externally accessible/controllable. This does not preclude the limited use of access panels which may be secured for flight while the vehicle is in the launch position.

SAF 2.3.3. All energetic device arming features shall be located on the airframe such that any inadvertent energy release by these devices will not impact personnel arming them.

For example, the arming key switch for an energetic device used to deploy a hatch panel shall not be located at the same airframe clocking position as the hatch panel deployed by that charge.

SAF 2.3.4. The arming mechanism should be accessible from ground level, without the use of ladders or other elevation devices, when the rocket is at a vertical orientation on the launch pad. In addition, each implemented arming device shall be of easy access, taking a maximum of 3 (three) seconds to do it.

2.4. Pressure Vessels

The following requirements apply to the design and verification testing of all pressure vessels, including combustion chambers; however, combustion chambers are exempt from the relief device requirement.

Burst Pressure for Metallic Pressure Vessels

SAF 2.4.1. Pressure vessels constructed entirely from isotropic materials (e.g., metals) shall be designed to a burst pressure no less than 2 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.

Burst Pressure for Composite Pressure Vessels

SAF 2.4.2. Pressure vessels constructed entirely from non-isotropic materials (e.g., carbon fiber, fiberglass, fiber-reinforced plastics), as well as metallic pressure vessels with composite overwrap (Composite Overwrapped Pressure Vessels — COPVs), shall be designed to withstand a burst pressure of no less than three (3) times the Maximum Expected Operating Pressure (MEOP).

The Maximum Expected Operating Pressure (MEOP) is defined as the maximum pressure expected during pre-launch, flight, and recovery operations.

Pressure Vessel Testing

The following requirements concern design and verification testing of pressure vessels.

SAF 2.4.3. Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location for pressure vessel testing.

SAF 2.4.4. Pressure vessels shall be proof pressure tested successfully to, at least, 1.5 times the maximum expected operating pressure for no less than twice the maximum expected system working time, using the intended or similar flight articles.

- The only acceptable type of test for this provision is the hydrostatic test.
- The maximum system working time is defined as the maximum uninterrupted time duration the vessel will remain pressurized during pre-launch, flight, and land operations.
- A full and detailed description of the used experimental methodology shall be included in the final report.

SAF 2.4.5. Teams should complete all tests at least three months prior to the test or launch window.

Relief Device

SAF 2.4.6. Pressure vessels shall implement a relief device, set to open at no greater than the proof pressure specified in the following requirements. Combustion chambers are exempted from this requirement.

SAF 2.4.7. In hybrid/liquid propulsion systems, the venting or pressure relief shall be directed outwards the rocket. No venting is allowed into confined or semi-confined volumes.

Section 3 — Propulsion System (PRS)

Section 5 defines the Propulsion System (PRS) for the experimental rocket projects, including types of propellants, ignition systems, and required tests.

3.1. Guidance

PRS 3.1.1. Teams shall utilize a liquid or hybrid researched and developed (R&D) propulsion system. An R&D propulsion system is defined as a propulsion system designed by the team, regardless of whether manufacturing is performed by the team or by a third party according to the team's design and specifications.

PRS 3.1.2. Multistage launch vehicles, side boosters, propulsion systems containing solid propellant, PET-bottles, water-based rockets, and toxic propellants are not allowed.

PRS 3.1.4. The minimum thrust-to-weight ratio for all mission categories shall be at least 1.25:1.

Thrust-to-weight ratio will be calculated based on either initial thrust of the motor or the average thrust of the motor (whichever is greater), divided by the takeoff weight (launch vehicle plus payload) of the experimental rocket.

PRS 3.1.5. Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s). The requirements concerning verification testing of researched and developed propulsion systems are detailed in the next provisions.

PRS 3.1.6. Teams should complete all tests 4 (four) months prior to the event. While not a requirement, this date is recommended to assure teams are prepared for the event.

3.2. Propellants

PRS 3.2.1. All propellants used shall be non-toxic. Nitrous oxide, kerosene, alcohol and similar substances, are all considered non-toxic.Toxic propellants are defined as those requiring breathing apparatus, special storage, transport infrastructure, extensive personal protective equipment (PPE), etc. (e.g., Hydrazine and N2O4).

PRS 3.2.2. Liquid oxygen (LOx), hydrogen peroxide and similar high oxidative fluids are prohibited and shall not be used due to its intrinsic risks, especially in an environment with high vegetation density.

3.3. Engineering Recommendations and Testing

PRS 3.3.1. Hybrid/liquid propulsion system shall implement a means for remotely controlled venting or offloading of all liquid and gaseous propellants in the event of a launch abort.

PRS 3.3.2. All hybrid/liquid combustion chambers shall be designed and tested according to the pressure vessel safety requirements defined in Section 2. Note that combustion chambers are exempted from the requirement for a relief device.

PRS 3.3.3. All propulsion systems using liquid or gaseous propellant(s) shall successfully (without significant anomalies) complete a propellant loading and off-loading test near as possible of the final "test or launch-configuration". Officials understand that tests may require additional or different setups for compliance to the location safety requirements.

Tests may be conducted using either actual propellant(s) or suitable proxy fluids. Links to videos and testing data shall be posted in your final report. A full and detailed description of the used experimental methodology shall be included in the final report.

PRS 3.3.4. While not a mandatory requirement, all propulsion systems should successfully (without significant anomalies) complete an instrumented (chamber pressure and/or thrust force), full scale (including system working time) static hot-fire test prior to the LASC.

This test does not need to be performed with the same tank parts, motor casing and/or nozzle components intended for use during the launch operations (e.g., teams shall verify their design but are not forced to design reloadable/reusable motor cases).

Fill Monitoring

PRS 3.3.5. Each hybrid/liquid propulsion system shall have a fill monitoring system that alerts when the tank(s) has reached the desired fill state, whether that is a partial fill or a full fill.

It is recommended that a calibrated cantilever load cell is used to accurately measure and transmit to the launch controller the actual fill state. Other methods can be used, but assuming that the condensation plume can be seen during the Summer Season in Brazil may be not acceptable by event officials.

There are simple methods such as using a temperature sensor. A capacitive sensor is also another reliable solution. Filling times and fill status detection must be tested prior to arrival at the event..

3.4. Propulsion Safing and Arming

PRS 3.4.1. The “arming action” is any action enabling an ignition signal to ignite the propellant(s).

PRS 3.4.2. A propulsion system is considered armed if only one action (e.g., an ignition signal) shall occur for the propellant(s) to ignite.
The "arming action" is usually something (i.e., a switch in series) that enables an ignition signal to ignite the propellant(s). For example, a software-based control circuit that automatically cycles through an "arm function" and an "ignition function" does not, in fact, implement arming.

In this case, the software's arm function does not prevent a single action (e.g., starting the launch software) from causing unauthorized ignition. This problem may be avoided by including a manual interrupt in the software program.

PRS 3.4.3. All propulsion system ignition circuits/sequences shall not be "armed" until all teams' launch personnel are at least 150 meters away from the launch vehicle.

PRS 3.4.4. The circuit arming shall be made by the event officials, after the Launch Area is clear, and only after all involved personnel are aware of this action.

PRS 3.4.5. Air-start ignition circuit arming is prohibited in any type of propulsion system designed for the mission categories of the Latin American Space Challenge.

Section 4 — Airframe and Aerodynamics (STR)

Section 6 defines the Airframe and Aerodynamics (STR) overall guidance and requirements for the development of experimental lander projects, including material selection and prohibitions.

The following provisions address some key points applicable to the experimental lander projects, but are not exhaustive of the conditions affecting each unique design. Each team is responsible for thoroughly understanding, analyzing, and mitigating their design's unique load set.

4.1. Overall Structural Integrity

STR 4.1.1. Lander vehicles shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling transportation, and the hop operations.

4.2. Material Selection and Tests

STR 4.2.1. Experimental landers should be constructed from lightweight materials such as fiberglass, carbon fiber, and/or ductile lightweight metals like aluminum. The choice of materials and construction techniques should be appropriate for the specific requirements of the planned flight to ensure optimal performance and safety.

STR 4.2.3. The use of heavy metals should generally be minimized for your team's own benefits to overall vehicle mass fraction, cost, vehicle mass, and safety in recovery or motor failure events. The use of composite materials such as fiberglass, kevlar, carbon fiber, et al. as appropriate are strongly suggested. The use of natural composites such as wood, cardboard, hemp fiber composites, bamboo composites, etc. should be examined for effectiveness and appropriateness.

STR 4.2.5. Teams should simulate and/or test the properties of materials and the strength of components developed for use in their experimental rocket projects. This process should include both individual and joint analyses to ensure the integrity of all parts.

It is important to verify the suitability and durability of all materials and parts under the specific conditions they will face during their missions, as this is essential for ensuring launch safety.

STR 4.2.6. Environmentally hazardous materials shall not be used in the experimental rocket project. This includes lead, mercury, uranium, and any other radioactive materials.

4.3. Best Practices and Recommendations

Load Bearing Eye Bolts and U-Bolts

STR 4.3.1. All load bearing eye bolts shall be of the closed-eye, forged type.

STR 4.3.2. All load bearing eye bolts, U-bolts, and links material shall be steel. This requirement extends to any bolt and eye-nut assembly used in place of an eyebolt. Stainless steel components (eye bolts, U-bolts, links, etc.) are permissible for use in recovery systems.

Joining of Airframe Sections

STR 4.3.3. Airframe joints shall be constructed to prevent bending. This ensures structural integrity under operational stresses.

STR 4.3.4. Coupler tubes used to join two airframe sections that are not intended to separate during flight shall extend a minimum of one-half caliber into each adjoining airframe section. Extension of one full caliber into each section is recommended to improve structural integrity.

Color Scheme and Markings

STR 4.3.5. All experimental anders shall be labeled with the Mission ID. The number assigned shall be clearly identified on the vehicle airframe, and prominently displayed, assisting officials to positively identify the project hardware with its respective team throughout the LASC. This label shall be duplicated on each part of the rocket which could separate either as designed or accidentally.

STR 4.3.6. Reference marks identifying the center of gravity and pressure shall be clearly placed on the airframe. These marks are essential for appropriate identification during the Launch Readiness Review (LRR), ensuring that all aspects of the lander's balance and aerodynamic stability are correctly assessed.

Section 5 — Electronics and Control Systems (ECS)

Section 7 defines the Electronics and Control Systems (ECS) guidance and overall requirements, including electronics and control systems.

5.1. Official Altitude Logging System

ECS 5.1.1. All experimental landers shall incorporate a telemetry system capable of recording and transmitting flight data. The telemetry system may be either a Commercial Off-The-Shelf (COTS) system or a team-developed researched and developed (R&D) system.

The following COTS telemetry systems are provided as recommended options:

Table 5: List of Approved Telemetry Systems.

Manufacturer	Model	Link
Featherweight	**GPS Tracker with Ground Station (Full System)**	Featherweight GPS Tracker
Multitronix	**TelemetryPro® Kate-3 Transmitter & Receiver**	Multitronix Products

Officials will check and validate data transmitted and received from the telemetry system during the Launch Readiness Review (LRR).

ECS 5.1.2. Transmission/reception frequencies shall be coordinated before the opening of the Sugarcane Launch Range Launch Area. Teams may be told to change frequency to a different channel if there is overlap. Final frequencies will be shared for all teams' awareness and will be displayed on a board at the event.

ECS 5.1.3. Teams shall be able to disarm any type of flight computer and/or telemetry system from the outside of the experimental lander while it is still on the pad. Flight computers and/or telemetry systems shall be armed once the lander is on the pad and before igniter leads are connected in accordance with Safety Policies.

5.2. Power Systems and Critical Wiring

On-Board Power Systems and Rail Standby Time

Loss of launch slots have been experienced on multiple occasions as onboard batteries are typically located in inaccessible positions. Despite the requirement of battery life on the launch pad, an unsuccessful operation attempt typically results in the teams deciding to:

- Disarm any energetic pyrotechnics;
- Take the experimental vehicle off the launch pad;
- Haul the experimental lander back to the team's preparation area;

- Use tools to perform medium to extensive disassembly of the experimental vehicle to extract batteries;
- Spend one to several hours recharging the batteries, if charged spares are not readily available;
- Perform the whole operation in reverse and return to the launch pad many hours later, to perform an additional launch attempt, if the possibility is given.

ECS 5.2.1. In order to overcome critically inefficient use of valuable and limited launch campaign time, including slot availability, teams should adopt one of the following two strategies:

- Implement an on-board charging and charge level maintenance system using an umbilical connection and cable; and/or
- Place all rechargeable or replaceable batteries conveniently under service panels accessible from ground level, without resorting to ladders, and having several spare sets of charged batteries ready at any time.

The implementation of an on-board charging and charge level maintenance system, based on a vehicle-wide charging bus and an umbilical cable, connected to a ground-based power supply, should be designed/implemented as follows:

- A "charging bus" should run along the entire length of the experimental vehicle, interfacing to all batteries to facilitate charging and continuous charging and subsequent maintenance trickle charging;
  - → Use mating connectors at every structural joint;
  - → Largely all benefits of the system are lost if even a single battery is left out of the umbilical charging bus system.
- Each tap-off from the on-board charging bus to individual battery subsystems shall be reverse current flow protected by a suitably rated diode;
  - → All on-board batteries should feature the same nominal voltage;
  - → If bus voltage step-down is required for batteries with lower nominal voltage, adequately heat-dissipated linear regulators are strongly recommended and placed upstream of the mandatory cell balancing circuits;
  - → LiPo battery cell balancing circuits of up to 12S cell count are widely available as pre-assembled PCBs for a low price, complete with built-in under voltage-cut-off, overcurrent-protection and overcharging cut-off;
  - → Vehicle batteries could all be considered "permanently" installed, not requiring removal past initial installation during on-site preparation. The ground-based power supply should simply be outputting the battery trickle charge voltage, plus a diode drop, for easiest implementation.

The advantages of implementing such a system are in most cases worth the effort. Most significantly, the experimental vehicle is always tested with 100% peak charged batteries.

Remove Before Flight (RBF)

ECS 5.2.2. The onboard batteries shall be connected to the electronics system via a "Normally Closed" switch or a Pull Pin Switch.

Given the risk of accidentally pulling the pin and thereby activating the electronics system, it is recommended to secure the end of the pin with a piece of masking tape. This precaution helps prevent unintentional removal and activation, ensuring the safety and stability of the system prior to intended use.

ECS 5.2.3. Teams shall attach Remove Before Flight (RBF) tags, flags, or protective covers to the following components:

- All pull-pin switches used in the experimental rocket
- All dynamic pressure ports or holes used for avionics
- The tip of the nosecone

In addition, teams should attach Remove Before Flight tags, flags, or protections to the following components

- Nozzle (Note: tags must not be fixed directly to the nozzle; they should be gently attached)
- All serviceable ports
- The leading edge of all fins
- Rail buttons

All Remove Before Flight (RBF) tags must be red with white lettering, displaying one of the following texts:

- “RBF”
- “Not for Flight”
- “Remove Before Flight”

Each tag should also include an attached label or identification tag with specific component details, including component name, and item number.

Safety Critical Wiring and Switch Requirements

Safety critical wiring is defined within this document as the electrical wiring that is involved with the deployment events of the recovery system. To facilitate the replacement of electronics in the event of a failure and to simplify assembly, teams should prioritize cable management.

Wiring and switches need to be securely fastened to prevent dislodging during launch and recovery operations. Given that experimental rockets will encounter high-G loads during boost phases and parachute deployment, which can displace wiring or toggle switches, it is crucial to secure these components firmly. Teams shall ensure that switches remain in the armed position

throughout the flight and do not accidentally deactivate during any flight events. This attention to detail in securing and managing wiring and switches is essential for maintaining the integrity and safety of the recovery system.

ECS 5.2.4. All safety critical wiring shall implement a cable management solution (e.g., wire ties, wiring, harnesses, cable raceways) which will prevent tangling and excessive free movement of significant wiring/cable lengths due to expected launch loads.

This requirement is not intended to negate the small amount of slack necessary at all connections/terminals to prevent unintentional de-mating due to expected launch loads transferred into wiring/cables at physical interfaces.

ECS 5.2.5. All safety critical wiring/cable connections shall be sufficiently secure as to prevent demating due to expected launch loads. This will be evaluated by a "tug test", in which the connection is gently but firmly "tugged" by hand to verify it is unlikely to break free in flight.

5.3. Attitude Control Systems (ACS)

Attitude Control Systems (ACS) requirements and procedures will be completely published at a later stage according to the official milestone roadmap and event schedule.

Designed to Fail Safe

ECS 5.3.1. Attitude Control Systems (ACS) shall mechanically lock in a neutral state whenever either an abort signal is received for any reason, primary system power is lost, or the launch vehicle's attitude exceeds 15° from its launch elevation.

Any one of these conditions being met shall trigger the fail safe, neutral system state. A neutral state is defined as one which does not apply any moments to the launch vehicle (e.g., aerodynamic surfaces trimmed or retracted, gas jets off, etc.).

Attitude Control System Electronics

ECS 5.3.2. Wherever possible, all ACS should comply with requirements and goals for "redundant electronics" and "safety critical wiring" as recovery systems. As for all electronics, it is highly recommended to ensure easy and quick access to switches/connectors via an access panel on the airframe. Access panels should be positioned so they are reachable from ground level, ideally without ladders. Access panels shall be secured for flight.

Section 6 — Hot-fire Test Operations (HFT)

Section 6 shall define the Hot-Fire Test Operations (HFT) procedures and establish the minimum requirements for the throttleable hot-fire test (Milestone 2) and the Thrust Vector Control (TVC) test (Milestone 3) conducted as part of the Latin American Space Challenge milestones.

This section will be published at a later stage according to the official milestone roadmap and event schedule.

Section 7 — Launch Operations (FLT)

Section 7 shall define the Launch Operations (FLT) procedures and specify the minimum technical, operational, and safety requirements applicable to the tethered flight test (Milestone 4) and the hop flight test (Milestone 5).

This section will be published at a later stage according to the official milestone roadmap and event schedule.

Section 8 — General Requirements (GEN)

Section 7 defines the General Requirements (GEN) and establishes the overall guidance and requirements for the development of experimental lander projects. The following provisions address key requirements applicable to all lander projects, regardless of mission category, within the Latin American Space Challenge.

8.1. Administrative Requirements

GEN 8.1.1. Teams are expected to actively collaborate with other teams participating in the competition. This includes regularly sharing project updates, engaging with the technical community, sharing CAD models, test results, and technical lessons learned when appropriate, and, by default, inviting and attending each other's design reviews. Teams are encouraged to foster a collaborative environment and remain open to sharing resources and technical knowledge when possible.

GEN 8.1.2. Teams shall agree to host LASC Sponsors and Partners for meet-and-greet events and facility tours upon request from the LASC organization.

GEN 8.1.3. Each mission shall display LASC Sponsors and Partners logos on propulsion systems, experimental landers, and other major hardware in a prominent location, as requested by the LASC organization.

Appendix A: Details for the Milestone Report

Abstract

The Milestone Report shall include an Abstract. At a minimum, the abstract shall identify the mission category in which the team is competing, describe any unique or defining design characteristics of the project, and provide any additional information necessary to convey the high-level goals and objectives of the project.

Introduction

The Milestone Report shall include an Introduction. This section shall provide an overview of the project, stakeholders, team structure, mission organization, and project management approach. The introduction may repeat some of the content included in the Abstract, as the Abstract is intended to function as a standalone summary when necessary.

Mission Concept and Proposed Roadmap

The Milestone Report shall include a Mission Concept and Proposed Roadmap section. This section shall begin with the motivation for developing the experimental lander project, followed by the proposed mission concept, a detailed roadmap covering all remaining milestones, and an identification of the proposed propulsion system.

System Architecture

The Milestone Report shall include a System Architecture Overview. This section shall present a top-level overview of the integrated system, including a cutaway or diagram depicting the fully integrated project and its major subsystems or modules configured for the proposed mission. This section shall also include a subsection specifically describing the proposed propulsion system.

Concept of Operations

The Milestone Report shall include a Concept of Operations for each milestone described in the proposed roadmap. This section shall identify the mission phases, include mission profile figures, and describe the nominal operation of all subsystems or modules during each phase (i.e., what occurs during each phase and which subsystems are responsible for each function).

Preliminary Budget

The Milestone Report shall include a Preliminary Budget for each milestone described in the proposed roadmap. This section shall identify the estimated capital investment and operational expenditures associated with each milestone in the proposed roadmap. A total program budget shall also be presented and discussed, including the expected sources of funding, sponsorships,

fundraising activities, and other resource acquisition strategies planned by the team to support the project throughout all milestones.

Weights, Measures, and Performance Data

The Milestone Report shall contain preliminary Weights, Measures, and Performance Data of the experimental lander project. This requirement will be satisfied by informing the data in this section of the document.

Lessons Learned and Next Steps

The Milestone Report shall include a Lessons Learned and Next Steps section. This section shall summarize lessons learned during concept development and design, including both team management and technical development aspects.

The section shall also define the next steps in the proposed roadmap. Additionally, this section should include strategies for knowledge transfer from senior team members to newer members who will assume future leadership and technical roles.

Hazard Analysis Appendix

The first Milestone Report appendix shall contain a Hazard Analysis Report. This appendix shall address as applicable, hazardous material handling, transportation, storage procedures, and any other aspects of the design which pose potential hazards to operating personnel. A mitigation approach – by process and/or design – shall be defined for each hazard identified.

Risk Assessment Appendix

The second Milestone Report appendix shall contain a Risk Assessment. This appendix shall summarize risk and reliability concepts associated with the project. All identified failure modes which pose a risk to mission success shall be recorded in a matrix, organized according to the mission phases identified by the ConOps.

A mitigation approach – by process and/or design – shall be defined for each risk identified. A common description of the Risk Assessment is FMECA (Failure Mode and Effect Criticality Analysis). A Risk Assessment/FMECA is often represented as a spreadsheet matrix. The input to the matrix is listed as follows:

- A description of the identified failure mode.
- The likelihood of the failure mode occurring.
- The severity and impact of the failure mode occurring

The likelihood of a failure mode occurrence and the severity of the occurrence is assigned values according to the following tables:

Table 8: Likelihood of Failure.

Failure Probability	Value	Assessment of Risk
Remote	1	This is unlikely to happen.
Occasional	2	This might happen.
Probable or likely	3	This is likely to happen.

Table 9: Severity of Occurrence.

Mishap Severity	Value	Effect of Failure Mode
Minor or negligible	1	Minor impact on mission.
Critical	2	Deterioration of performance and mission.
Catastrophic	3	Safety hazard and/or likely loss of mission

The "Criticality Ranking" is the product of the Failure Probability and the Mishap Severity. The criticality rating is a measure of how urgent and how severe mitigation actions will have to be taken, to reduce the Criticality Ranking.

Table 10: Criticality Ranking.

Mishap Severity	Value	Effect of Failure Mode
1	Minor	This failure mode is not a concern.
2	Minor	This failure mode is of very minor concern.
3	Medium	Justification needed. Jury may decide to review.
4	High	Technical jury approval needed before launch.
6	Critical	Action required to reduce ranking before launch.
9	Critical	Action required to reduce ranking before launch.

The output of the matrix is highlighting and ranking failure mode liabilities to the mission, and the justifications and mitigations to reduce the Criticality Ranking. An example is given below:

Table 11: Risk Matrix.

Failure Mode	Mission Phase	Failure Probability	Mishap Severity	Critically Ranking	Mandatory Comments
Battery Thermal Runaway	Pre-launch phase	1	3	3	A thermal runaway begins when the heat generated within a battery exceeds the amount of heat that is dissipated to its surroundings. A battery balancing system will be implemented to reduce the risk.

Antenna Deployment failure	Deployment phase	2	3	6	Antenna deployment is mandatory for communication. A redundant system is required.

A typical FMECA scale for the complexity of projects attending LASC should feature no less than 10 identified, ranked, commented, and justified failure modes – these should address at the minimum all important and critical failure modes.

Engineering Drawings and Optional Appendix

The final Milestone Report appendix shall contain Engineering Drawings and other optional appendices. This appendix shall include any revision controlled technical drawings necessary to define significant subsystems or components.

Appendix B: Detailing Grading Criteria

The grading will be conducted by the event officials based on the individual grading criterion in the respective mission categories. A summary and overview of the grading scheme is given below for clarity.

Table 12: Grading Scheme.

Count	For Countable/Relative Criteria
91 to 100%	Outstanding Quality/Conformity
76 to 90%	High Quality/Conformity
51 to 75%	No Greater than Average
Up to 50%	Unsatisfactory

Count	For Absolute Criteria
100%	Yes
0%	No

This is meant to be an intuitive and transparent scheme for the jury to follow and the teams to understand.

Team Effort

Team Effort will be graded up to 250 points (25% of 1,000 points possible). The total points for Team Effort is a combination of countable, relative and absolute criteria.

Table 13: Team Effort Grading Criteria.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
1st Progress Update	25 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	**/25**

Criterion	Outstanding	High	Average	Unsatisfactory	Points
2nd Progress Update	25 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	**/25**
3rd Progress Update	25 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	**/25**
4th Progress Update	25 points Compliant to Section 1.4.	-	-	0 point One or more required items missing.	**/25**
Mission Patch	46-50 points Completely complies with guidance. Excellent quality and design.	38-45 points Complies with the guidance with a few design or quality issues.	25-37 points Minimally complies with guidance. Medium quality and design.	<25 point Does not comply with guidance.	**/50**
Pitch Video	90-100 points Completely complies with guidance. Excellent quality, & clarity,	76-90 points Complies with the guidance with a few minor issues. High video and/or sound quality.	50-75 points Minimally complies with guidance. Medium video and/or sound quality. < 10% over or < 20% under time limit.	< 50 points Does not comply with guidance. Low video and/or sound quality. > 10% over or > 20% under the time limit.	**/100**
Team Effort					**/250**

Milestone Report

The Milestone Report will be graded up to 400 points (40% of 1,000 points possible) divided into three criteria: Completeness (40 points), Correctness (up to 80 points) and Analysis (up to 280 points). The Completeness Criterion will be scored using Absolute Criteria. The Correctness and Analysis Criteria will be scored in a mix of Countable and Relative Criteria.

Table 14: Milestone Report Grading Criteria - Completeness Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Completeness	40 points All required items of the Appendix A present.	-	-	0 point One or more required items missing.	**/40**

Table 15: Milestone Report Grading Criteria - Correctness Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Style	36-40 points Writing was exceptionally clear, understandable, and concise. Sentence and paragraph organization is exceptional. Writing is free of	30-35 points The writing was clear, understandable, and concise. Overall paragraph and sentence organization were very good. Digressions or irrelevant information do not significantly	20-29 points Writing was generally clear and understandable. Paragraph and sentence organization were generally good. Digressions or irrelevant information detract from the	< 20 points Writing was repeatedly unclear, difficult to understand or wordy. Overall paragraph and/or sentence organization were ineffective or nonexistent. Digressions and/or irrelevant information	**/40**

	digressions or irrelevant information.	detract from the report	report's analysis.	consistently detract from the analysis.	
Mechanics	18-20 points No grammar, spelling, or mechanics errors. Scientific terms correctly used, units and dimensions consistent and correct.	12-17 points No more than a few grammar, spelling, or usage errors. Only a few minor errors with use of scientific terms or dimensions.	5-11 points Significant spelling, usage, and grammar errors that did not detract from readability. Significant errors with use of scientific terms or dimensions.	< 5 points Repeated grammar or spelling errors detracted from readability. Errors with use of scientific terms or dimensions detracted from the report.	**/20**
Format	18-20 points Completely follows the required template. Meets page limits.	12-17 points Minor deviations from required template. Meets page limits.	5-11 points Major deviations from required template. < 10% over page limits.	< 5 points No attempt at cohesive format or use of required template. More than 10% over page limits.	**/20**
Correctness					**/80**

Table 16: Milestone Report Grading Criteria - Analysis Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Depth of Analysis	90-100 points Very complete and thorough analysis. All key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	76-89 points Adequate analysis with minor weaknesses. Most key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	50-75 points Adequate analysis with significant gaps or weaknesses. Some key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs. Some minor incorrect statements.	< 50 points Inadequate analysis. Few, if any key design decisions were discussed No discussion of tradeoffs. Parts of analysis conflict with general scientific knowledge.	**/100**
Assumptions and Sensitivity Analysis	51-60 points All assumptions are clearly stated. Sensitivity analysis is performed to quantify uncertainty in variables and assumptions.	36-50 points Most assumptions were addressed. Some sensitivity analysis.	25-35 points Unstated assumptions. No sensitivity analysis.	< 25 points No stated assumptions or assumptions were unreasonable. No sensitivity analysis.	**/60**
Proposed Roadmap and Budget	71-80 points The proposed roadmap is clear, detailed, and logically structured across all milestones. Budget estimates are complete and realistic for each milestone.	51-70 points The proposed roadmap is mostly clear and includes budget estimates for the major milestones. Most verification and validation activities are adequately described.	30-50 points The roadmap and budget are partially defined or lack sufficient detail. Some verification and validation activities are described but not consistently.	< 30 points The roadmap and budget are unclear, incomplete, or unrealistic. It is unclear whether verification and validation activities are planned or performed.	**/80**
Use of Charts and Figures	31-40 points Tables, figures, and appendices all effectively organize and communicate information.	21-30 points Use of tables, figures, and appendices is mostly effective.	10-20 points Use of tables, figures, and appendices is somewhat effective with significant issues.	< 10 points Tables, figures, and appendices were incorrect or misleading.	**/40**
Analysis					**/280**

Then, the *total score* of the Milestone Report will be the sum of the points for Completeness, Correctness and Analysis.

Presentation Session

The Presentation Session will be graded up to 350 points (35% of 1,000 points possible) divided into two criteria: Presentation Materials (100 points) and Oral Presentation (250 points).

Table 14: Presentation Session Grading Criteria - Presentation Materials Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Completeness	20 points The Presentation Materials are complete and were submitted on time.	-	-	0 point One or more major required items missing.	**/20**
Mechanics	13-15 points No grammar, spelling, or mechanics errors. Scientific terms correctly used, units and dimensions consistent and correct.	8-12 points No more than a few grammar, spelling, or usage errors. Only a few minor errors with use of scientific terms or dimensions.	5-7 points Significant spelling, usage, and grammar errors that did not detract from readability. Significant errors with use of scientific terms or dimensions.	< 5 points Repeated grammar or spelling errors detracted from readability. Errors with use of scientific terms or dimensions detracted from the report.	**/15**
Format	13-15 points Completely follows the required template. Meets page limits.	8-12 points Minor deviations from required template. Meets page limits.	5-7 points Major deviations from required template. < 10% over page limits.	< 5 points No attempt at cohesive format or use of required template. More than 10% over page limits.	**/15**
Use of Charts, Figures and Videos	41-50 points Tables, figures, videos, all effectively presented.	31-40 points Use of tables, figures, videos is mostly effective.	20-30 points Use of tables, figures, videos is somewhat effective with significant issues.	< 20 points Tables, figures, and videos were incorrect or misleading.	**/50**
Presentation Materials					**/100**

Table 17: Presentation Session Grading Criteria - Oral Presentation Criterion.

Criterion	Outstanding	High	Average	Unsatisfactory	Points
Presentation Session Performed	50 points Completely complies with Section 1.4.	-	-	0 point Does not comply with Section 1.4.	**/50**
Mission Concept, Roadmap and System Engineering	91-100 points A clearly understood design vision with achievable goals, aligned with strategic objectives, and strong evidence of systems engineering discipline throughout the	76-90 points Design vision is mostly understood and achievable with coherent goals. Key project elements address team goals. Good systems engineering discipline throughout development.	50-75 points Design vision is incomplete or questionably achievable. Unclear how project elements address team goals. Some lapses in systems engineering discipline. Inconsistent support for team goals and	< 50 points Questionable or unachievable design vision. Most project elements do not address team goals. Major lapses in systems engineering discipline. No unified design goals,	**/100**

Criterion	Outstanding	High	Average	Unsatisfactory	Points
	design team.	Most of the design team supports coherent and understood goals.	some evidence of design team working at cross-purposes.	with clear evidence of team members working at cross-purposes.	
Depth of Analysis	46-50 points Very complete and thorough analysis. All key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	38-45 points Adequate analysis with minor weaknesses. Most key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs.	25-37 points Adequate analysis with significant gaps or weaknesses. Some key design decisions are discussed and based on design targets, constraints, and appropriate tradeoffs. Some minor incorrect statements.	< 25 points Inadequate analysis. Few, if any key design decisions were discussed No discussion of tradeoffs. Parts of analysis conflict with general scientific knowledge.	**/50**
Team Knowledge	46-50 points Strong team understanding of the principles governing design and reasoning behind the design. All members of the team can clearly articulate reasoning for choices.	38-45 points Generally good team understanding of the physical principles governing design and reasoning behind the design. Team members defer to a few team "experts" during discussion.	25-37 points Some team understanding of the physical principles governing design and reasoning behind the design. Team members defer to one or two team "experts" during discussion.	< 25 points Inadequate team understanding of the principles governing design and reasoning behind the design.	**/50**
Oral Presentation					**/250**

Final Score

The maximum points possible are up to 1,000 points, including Team Effort, Milestone Report and Presentation Session.

Final Score = (Team Effort + Milestone Report + Presentation Session) - Penalties